

ReproBot: A Multi-Agent System for Automated Scientific Paper Replication

Second Progress Report

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Abstract

The first progress report described ReproBot as a proposed four-agent pipeline for automated machine learning paper replication, supported by a literature review, an implementation-ready project plan, and an early Reader prototype. This report covers the implementation phase. Five of the seven planned components are now built and verified by execution: PDF extraction, a five-field structured Reader, a Coder that emits a self-contained training script, a Docker-sandboxed Runner, and an Orchestrator owning the shared-memory state and the Coder $\leftrightarrow$ Runner retry loop. Two papers – *Network In Network* and *Wide Residual Networks* – have been carried end to end, from PDF through to a training script executing inside a container and reporting metrics back. We report three substantive findings. First, the retry loop repairs real defects: a deliberately reintroduced runtime fault was triaged, fed back to the Coder as targeted feedback, and repaired on the first retry. Second, structured architecture extraction measurably outranks the language model’s pretrained priors – given *Network In Network*, whose contribution is a custom `mlpconv` layer with no memorized reference implementation, the Coder produced the correct layer cascade and global-average-pooling head rather than the generic convolutional stack its priors would favour. Third, several failure modes in LLM tool use are silent by construction and require deterministic guards rather than prompt engineering. We are explicit about what remains: the Critic does not exist, so no reproduced number has yet been compared against any paper’s claim, and the measured CPU throughput places a full-fidelity training run at roughly 22 days per paper, making GPU access a precondition for any fidelity result.

1 Introduction

The first progress report [1] established ReproBot’s motivation and design. The problem it addresses is unchanged: reproducing a machine learning paper’s empirical results is a mechanical task in principle and a labor-intensive one in practice, and the large majority of published work ships no usable implementation – across ICLR, ICML and NeurIPS 2024, only 16.7–21.2% of accepted papers included a working one [3]. The gap ReproBot targets is equally unchanged: no surveyed system combines parsing a paper’s claims, executing a faithful implementation, and explicitly verifying the reproduced numbers against the published ones with a repair loop driven by that comparison.

What has changed is that the system now exists in code. This report is therefore structured around evidence rather than intent. Section 2 describes the architecture as implemented, and where it departs from the plan. Section 3 documents each stage. Section 4 reports the measured results of running two papers through the full pipeline. Section 5 collects engineering findings that we believe generalize beyond this project. Section 6 states the limitations plainly, including the ones that prevent us from making any replication-fidelity claim at all. Sections 7 and 8 cover next steps and conclusions.

Since the first report, the project has produced 30 commits across five newly implemented pipeline stages totalling approximately 6,700 lines of Python, together with 21 logged delegations to specialized coding, research, review and validation subagents.

2 Implemented Architecture

The pipeline follows the planned Reader $\rightarrow$ Coder $\rightarrow$ Runner shape, with two departures worth stating.

The Reader emits five fields, not four. The project plan specified `method_summary`, `claims`,

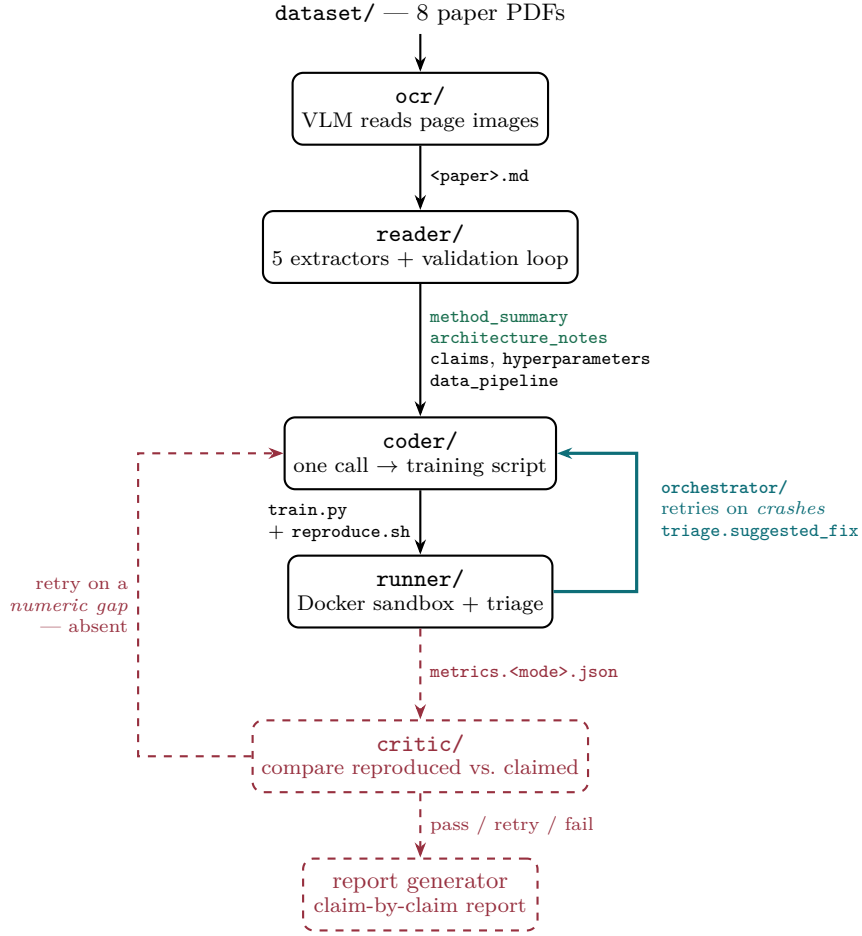


Figure 1: ReproBot as implemented. Solid components are built and verified by execution; dashed components in red do not exist. Two loops are shown and only one is real: the Orchestrator retries on execution failures, routing the Runner’s triage back to the Coder as feedback. The absent loop — retrying on a numeric gap between the reproduced value and the paper’s claim — is the one the literature review identifies as missing from every surveyed system, and it is missing here too.

`hyperparameters` and `architecture_notes`. All four are implemented. A fifth, `data_pipeline`, was added on the plan’s own recommendation: data-processing detail is the single weakest-covered aspect of paper-to-code extraction across every surveyed system (56% coverage in PaperCoder against 79–92% elsewhere [3]), and is therefore treated as a first-class output field rather than folded into the method summary.

The Runner’s interface is a generated shell script, not a command line. The Coder emits a `reproduce.sh` alongside every training script, exposing four modes — `probe`, `smoke`, `capped` and `full` — and the Runner invokes nothing else. It never constructs a Python command and never passes a flag. This keeps the Runner paper-agnostic: a future paper whose script requires entirely different arguments changes only its own `reproduce.sh`. The contract narrows from eight exactly-spelled command-line flags to four mode names.

The Orchestrator is implemented as a bounded, deterministic Python loop rather than the Lang-Graph state machine the plan proposed. At the present scope the graph would comprise three nodes and one conditional edge, which does not justify the dependency. The shared-memory object follows the plan’s schema exactly and the transition logic is isolated in pure functions, so migration remains mechanical once the Critic introduces genuine branching.

3 Stage Implementations

3.1 Extraction: ocr/

Four independent PDF-to-Markdown backends share one command-line shape. The vision-language backend renders each page to an image and sends it to Claude, so figures are genuinely read rather than inferred from surrounding text — a property the Reader’s architecture extraction depends on directly (Section 3.2). Six of the eight target papers are extracted. Two fail on a deterministic content-filter false positive, diagnosed as originating in the vision classifier rather than in our prompt: the same page fails identically at different render scales and under a minimal one-line prompt.

3.2 Structured extraction: reader/

Five extractors implement a shared abstract base class, so the pipeline treats every stage uniformly and a new extraction type requires one new class and one line in the stage list. A validator makes one further model call over the combined output and the paper text, flagging inconsistencies without fixing them; the pipeline routes each flag back to the single stage its `relates_to` prefix names, re-runs that stage with the flag folded into its prompt as feedback, and re-validates, capped at three passes.

Two properties of `architecture_notes` proved load-bearing.

Recording what the paper does not say. The field `unstated_details` is required, not optional. *Network In Network* describes three `mlpconv` layers and a global average pooling layer in prose and in its Figure 2, but never states filter counts, kernel sizes or strides in the main text, deferring them to supplementary material. The extractor records twelve such gaps, quoting the paper’s own deferral, and invents no values. This matters more here than elsewhere in the pipeline: code has to run, so a confidently invented channel count does not fail loudly — it trains the wrong network and reports a number for it.

Distinguishing the paper’s own equations from its contrasts. A paper introducing a new layer typically prints the conventional formulation, and often a rival’s, beside its own. *Network In Network* prints all three within two pages. Returned as an undifferentiated list, these are three plausible specifications and a code generator will implement one of them. Instructing the model to omit baseline equations did not work; it continued to return them. Requiring the *schema* to carry an `is_own_method` flag on every entry did work, and the distinction survives into the generated code.

3.3 Code generation: coder/

One model call converts a paper’s Reader output plus its Markdown into a self-contained HuggingFace `Trainer` script. Input precedence is explicit and ordered: `architecture_notes` first, the paper text as corroboration, and the model’s own recollection last and only when disclosed in the output’s `assumptions` field.

Two deterministic gates run before a script is accepted, both free of model calls. The script must parse as valid Python, and every flag the Runner depends on must literally appear in its text. Neither gate can establish that the script *runs* — a limitation that recurs in Section 5.

3.4 Sandboxed execution: runner/

The generated script executes inside a Docker container built from a pinned CPU-only image. Stages escalate and stop at the first failure. Timeout enforcement required care: terminating the client process leaves the container running under the daemon, so every run is launched with an explicit name and killed by that name on expiry.

A single cheap model call triages a genuine failure into *recoverable* — a defect in the generated script — or *environmental*. Success and timeout are classified mechanically with no model call at all, which both saves cost on the common path and avoids inviting a judgment where the exit status already answers the question.

3.5 The retry loop: orchestrator/

The Orchestrator owns the shared-memory object and the loop. Its routing order is deliberate: an environmental error stops immediately rather than consuming retries, because regenerating a script cannot repair a broken container. A timeout likewise stops, since no triage is produced for timeouts by design and a slow but correct script must not be “repaired” into a different one. Only a recoverable error regenerates.

A plateau guard compares each regenerated script against its predecessor and stops early when they are near-identical, rather than exhausting the budget on a fix that is not changing anything — following PaperBench’s observation that agents plateau rather than making steady long-horizon progress [2].

4 Results

4.1 Extraction coverage

Table 1 reports the Reader’s output across the four papers processed. The *own/eq* column is the most informative: it counts equations marked as the paper’s own against the total captured. *All Convolutional Net* prints six equations of which two are its own; *Wide Residual Networks* prints one and it is its own. A field that always returned every equation would be useless to a code generator.

Table 1: Reader output across four papers. *Comp.* is architecture components; *own/eq* is own-method equations over equations captured; *Gaps* is `unstated_details` entries; *Flags* is unresolved validation flags after the three-pass cap.

Paper	Claims	Hyp.	Data	Comp.	own/eq	Gaps	Flags
Network In Network	12	17	4	7	10	1	0
All Convolutional Net	17	13	4	10	10	1	0
Deep Residual Learning	66	28	6	10	10	1	0
Wide Residual Networks	61	20	4	11	11	1	0

Validation does not converge. Every paper finishes with flags remaining after the cap, and the count is worst on the paper with the most claims. Inspection shows a substantial proportion are the validator raising a concern and dismissing it within its own description rather than identifying a real defect. We regard this as a genuine limitation of the current design rather than a tuning problem, and record it as such.

4.2 End-to-end execution

Both papers with generated scripts were carried through the Orchestrator. Both succeeded on the first attempt with no retries required (Table 2).

The roughly sixfold difference in wall clock reflects model size: WRN-28-10 carries approximately

Table 2: End-to-end runs. Both papers reached **smoke** and returned **success** with zero retries consumed. Reproduced values are reported to demonstrate that the measurement path works end to end; they are *not* replication results (Section 6).

Paper	probe	smoke	Reproduced	Claimed
Network In Network	32.2 s	62.1 s	86.72%	10.41%
Wide Residual Networks	241.4 s	367.2 s	89.45%	4.00%

36.5M parameters against *Network In Network*’s order of 1M. The reproduced error rates are far from the claimed values, and this is the expected and correct behaviour of the stages that produced them — **probe** and **smoke** train for one epoch over a few hundred images and exist to answer whether the script runs, never whether the number is right.

4.3 The retry loop repairs real defects

The loop was validated against a deliberately reintroduced fault: a learning-rate scheduler whose method reads an attribute its constructor never assigns. The fault is valid Python, so the syntax gate admits it, and it raises only when the training loop first logs the learning rate. This is precisely the class of defect the loop exists for.

The container run failed with an `AttributeError`; triage classified it as recoverable and produced a targeted fix naming the class and the missing assignment; the Coder regenerated with that text folded into its prompt; the plateau guard measured a similarity of 0.41 against the previous attempt, well below the 0.98 stopping threshold; and the second attempt passed. The repair was not incidental — the regenerated script retained the class the triage named and added exactly the missing assignment. This is the local counterpart of AutoReproduce’s

ablation, in which removing the debug loop barely moved an LLM-judged alignment score while nearly tripling the performance-gap metric [4]: code can read as a faithful implementation and still be wrong if it is never executed.

4.4 Architecture extraction outranks pretrained priors

The Coder was verified against *Network In Network* rather than *Wide Residual Networks*, deliberately. A wide residual network is among the most thoroughly memorized architectures in the literature and would be reproduced correctly from priors alone, demonstrating nothing about the extraction. *Network In Network* is the honest test: its `mlpconv` layer is the paper’s contribution, its dimensions appear nowhere in the paper, and its equation list contains two decoys.

The generated script contains a genuine `mlpconv` cascade — a $k \times k$ convolution followed by two 1×1

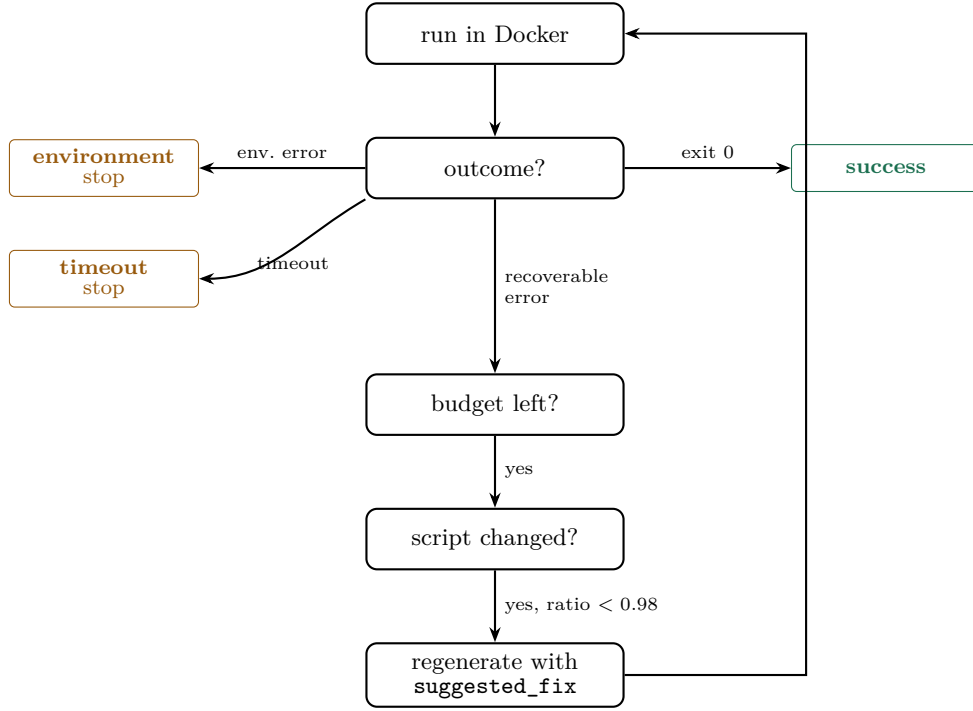


Figure 2: The Orchestrator’s routing. The order is deliberate: an environmental error stops *before* the retry budget is consulted, because regenerating a script cannot repair a broken container. A timeout also stops rather than retrying — no triage is produced for timeouts by design, and a slow but correct script must not be “repaired” into a different one. Only a recoverable error, within budget, whose regenerated script actually differs from its predecessor, results in another execution.

convolutions, with a docstring mapping each line to the paper’s Equation 2 — and contains no fully-connected layer at all, retaining the global-average-pooling classifier that is the paper’s second contribution. Both decoy equations are explicitly excluded and named as excluded.

The strongest evidence is negative. A priors-driven implementation of this paper is a plain convolutional stack with a fully-connected head; both of those defaults were avoided, and the generated script justifies the avoidance by citing components tagged [baseline, not own method] — a label from our extraction schema, not language from the paper.

We qualify this in one respect. The channel widths in the generated model are the canonical configuration, drawn from the model’s own knowledge rather than from the paper, which genuinely omits them. This is the designed outcome and it is disclosed in the output. The change made those values *visible* rather than *correct*, and a Critic will need to weight a disclosed prior below a paper-sourced value.

5 Engineering Findings

5.1 Silent failure modes in structured tool use

Three distinct failure modes were observed in which a model call returns a structurally valid response that is nonetheless empty or wrong, in a way indis-

tinguishable from a correct extraction over a paper that had nothing to report.

The first is output truncation at the token limit, which yields a partial payload. The second is an intermittent well-formed response with every required field absent. The third is the most instructive: a *double-encoded* payload, in which the model returns a clean tool-use response whose single key holds the entire result re-encoded as a JSON string. Field lookups then all miss and the extraction is discarded. It reproduced three times out of three on one paper, and fired again during later verification, where without the guard it would have reported zero datasets on an apparently clean run.

All three are now handled in one shared module rather than at six call sites. The general lesson is that structured-output failures do not necessarily announce themselves as errors, and a pipeline that treats a well-formed empty response as a legitimate negative result will silently accumulate them.

5.2 Schemas constrain models more reliably than instructions

Two independent cases: instructing the model to omit baseline equations did not stop it returning them, whereas requiring a per-entry `is_own_method` flag produced correct classification immediately; and a bookkeeping field the model never populated, despite being schema-required on every run, was elimi-

nated by deriving it from data already present rather than by asking more insistently. Where a value is computable from what has already been extracted, computing it is more reliable than requesting it.

5.3 Static gates cannot establish that generated code runs

Two defects reached generated scripts that passed every deterministic gate. The first is the scheduler fault described above. The second, found by inspection, is a divergence between a script’s own bookkeeping — which describes one stage as $192 \rightarrow 192 \rightarrow 10$ — and the code it emitted, which builds $192 \rightarrow 10 \rightarrow 10$. The script runs, and the choice is a defensible reading of a dimension the paper never states; what is wrong is that the record describes a different network from the code.

Neither is reachable by parsing. Both argue for the execution-driven loop rather than for additional static analysis.

5.4 Platform constraints shaped the architecture

The development machine is an Intel Mac, and no PyTorch release supports both Python 3.13 and Intel macOS: 3.13 support arrived in the same release range that dropped Intel-macOS wheels. Reducing to Python 3.11 installs PyTorch 2.2.2, but current **transformers** requires 2.5 or newer and, on finding an older version, silently disables its PyTorch backend rather than raising.

This is why the Runner executes inside a container rather than on the host. The container runs its own Linux Python, so the constraint does not apply within it, and the project’s dependency lock stays free of PyTorch entirely.

6 Limitations

We state these directly, because several bear on how the results above should be read.

No fidelity claim is possible yet. The Critic does not exist. Nothing in the system compares a reproduced number against a paper’s claim, and the field reserved for that verdict is permanently null. The reproduced error rates in Table 2 demonstrate that the measurement path works; they are not replication results and should not be read as any.

Full-fidelity runs are not currently feasible. Measured throughput is 5.256 training samples per second for WRN-28-10 on this CPU. The paper’s setup is 50,000 samples over 200 epochs, placing one paper and one claim at approximately 22 days. Every result reported here comes from stages that train for one epoch over a few hundred images. GPU access is a precondition for any fidelity measurement, not an optimization.

The evidence base is two papers. Both succeeded on the first attempt, so the retry loop’s behaviour in production remains demonstrated only by the injected fault. Four of the loop’s seven terminal verdicts have been exercised only against test doubles.

Retries regenerate rather than patch. A repair rewrites the whole file, so a successful fix can introduce a new defect elsewhere, and nothing compares the semantics of successive attempts. AutoReproduce’s line-range edit mechanism [4] is the natural remedy and is not yet implemented.

Validation does not converge, as reported in Section 4, and two of the eight target papers cannot currently be extracted at all.

7 Next Steps

The Critic is the highest-value remaining component and closes the loop the project’s thesis rests on. Its verdict logic should remain simple, auditable arithmetic against an explicit numeric tolerance rather than a model judging numbers by eye — Agent Laboratory’s automated reviewer overestimated quality by 2.3 points against human judges [6], and an objective numeric check is precisely the failure mode ReproBot exists to avoid. It can be built and tested against synthetic metrics immediately; its verdicts become meaningful when compute allows a real run.

GPU compute should be secured in parallel rather than afterwards, since it gates every fidelity result.

Coverage should be extended across the remaining papers, which requires no new code and would convert single demonstrations into distributions.

The Report Generator is the promised deliverable and the cheapest remaining component: the shared-memory object already carries everything it needs, including the append-only history log.

8 Conclusion

ReproBot has moved from design to a working pipeline. Five of seven components are implemented and verified by execution; two papers have been carried from PDF to a training script running in a sandbox; the retry loop repairs real defects; and structured architecture extraction demonstrably outranks the language model’s priors on a paper where those priors would otherwise dominate.

What has been demonstrated is that the machinery works end to end. What has not been demonstrated — and cannot be, until the Critic exists and compute allows a full training run — is replication fidelity, which is the project’s actual claim. We regard stating that distinction clearly as more useful than reporting an aggregate that would obscure it.

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